

Udit Deo

Ph.D. Scholar in Adaptive Visual Recognition under Distribution Shift
Indian Institute of Technology Roorkee

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Profile

Ph.D. scholar at the Machine Vision and Intelligence Lab, IIT Roorkee, investigating reliable and compute-efficient visual recognition under distribution shift. My current research focuses on adaptive inference in vision transformers—specifically, how calibration degradation under domain shift undermines confidence-guided computation, and how lightweight shift-aware mechanisms can restore allocation reliability for both image classification and dense prediction tasks. Previously, my work addressed fuzzy relation inequalities and multi-objective optimization frameworks for decision-making under uncertainty.

Research Experience

- 2025–Present **Reliable Adaptive Computation under Distribution Shift**
Machine Vision and Intelligence Lab, IIT Roorkee
- Studying confidence-guided adaptive computation in vision transformers, examining its reliability under synthetic corruptions (ImageNet-C) and real-world adverse conditions (ACDC).
 - Identifying how calibration degradation under distribution shift undermines confidence-based compute allocation, leading to unreliable token-level decisions.
 - Designing lightweight, shift-aware mechanisms that combine prediction confidence with feature-space divergence from clean-data representations to restore allocation reliability.
 - Evaluating adaptive token merging for image classification and extending the framework to dense prediction via SegFormer-based semantic segmentation.

Education

- 2025–Present **Indian Institute of Technology Roorkee**
Ph.D. in Computer Science and Engineering, GPA: 8.14/10
Advisor: Prof. R. Balasubramanian. Research area: reliable and efficient computer vision under distribution shift.
- 2023–2025 **Maulana Azad National Institute of Technology Bhopal**
M.Tech. in Software Engineering, GPA: 9.53/10
Secured 2nd rank in the program. Thesis: *Bridging Fuzziness and Optimization: Frameworks for Relation Inequalities and Multi-Criteria Decision Systems*.
- 2017–2021 **Shri Mata Vaishno Devi University**
B.Tech. in Computer Science and Engineering, GPA: 8.21/10
Received a tuition-fee waiver for securing 2nd rank in the seventh semester.

Publications

- 2025 **Robust Client–Server Quality Allocation via Ordered Weighted Averaging: A Multi-Objective Optimization Framework**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Computing, Communication and Networking Technologies (ICCCNT).
- 2025 **A New Approach to Fuzzy Relation Inequalities: Effects of Variable Absence and Weighted Composition**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Artificial Intelligence and its Application (ICAIA), pp. 127–138. doi: 10.1007/978-981-95-0493-0_10.
- 2025 **Advances in Fuzzy Relation Inequalities and Optimization Techniques**
Udit Deo, J. K. Jain, and M. M. S. Beg
International Conference on Computational Intelligence, Communication Technology and Networking (CICTN), pp. 833–838. doi:10.1109/CICTN64563.2025.10932336.

Awards & National Examinations

2024	UGC Junior Research Fellowship <i>Computer Science and Applications</i> 99.95 percentile.
2025, 2022	Graduate Aptitude Test in Engineering <i>GATE CS & DA</i> Qualified GATE CS in 2025 and 2022, and GATE DA in 2025.
2025	Academic Merit Distinction <i>MANIT Bhopal</i> Received a Merit Certificate at the 22nd Convocation for outstanding academic performance in the M.Tech. program.

Professional Experience

2021–2023	Amdocs Development Center <i>Associate Software Engineer, Gurugram</i> – Developed Python-based data-synchronization tools that reduced associated manual workload by approximately 90%. – Designed database comparison and merging workflows to improve consistency across systems.
2020	Stackfusion <i>Machine Learning Intern, Gurugram</i> – Fine-tuned MobileNetSSD, ResNet, and YOLOv3 models for vehicle axle and licence-plate recognition. – Developed Python-based preprocessing pipelines to improve training-data quality and diversity.

Patent

2025	A Device for Producing Electricity Using Roof Air Ventilator <i>Granted Design Patent, IP India</i> Design No. 440258-001, Class 13-01. Co-inventors: Anand Kumar, Ayush Kumar Agrawal, Uday Deo, Udit Deo, Shashank Kumar Soni, Bikramaditya Chakraborty, and Basundhara Singhdeo.
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Technical Profile

Research Areas	Reliable Computer Vision; Adaptive Computation; Distribution Shift; Model Calibration; Vision Transformers; Semantic Segmentation; Efficient Deep Learning.
Programming	Python; C++; SQL; Shell scripting.
Frameworks	PyTorch; TensorFlow; OpenCV; Scikit-learn; NumPy; Pandas; MATLAB.
Methods	Experimental evaluation; uncertainty analysis; feature-space analysis; statistical modelling; multi-objective optimization.
Tools	Linux; Git; LaTeX; Flask.